

Decay Alone Does Not Support Directional Wavelet–Riesz Interpolation

A counterexample to the open problem in arXiv:1409.2143

Automated mathematics run `fa_banach_001`, lane 2

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Status. Candidate counterexample, likely valid, pending human review. The result gives a full negative answer to the literal “decay estimates only” formulation. If direction-specific sectional oscillation is intended to remain implicit, the example instead identifies a sharp missing hypothesis.

1 Source problem

Muller and Muller [1] study estimates controlling a directional wavelet projection $W^{(\varepsilon)}$ by a Riesz transform R_{i_0} when $\varepsilon_{i_0} = 1$. Their wavelets are indexed by dyadic cubes $Q \subset \mathbb{R}^n$ and orientations $\varepsilon \in \{0, 1\}^n \setminus \{0\}$. The localization and decay condition (1.2a) is

$$|\varphi_Q^{(\varepsilon)}(x)| \leq C \left(1 + \frac{\text{dist}(x, Q)}{s(Q)} \right)^{-n(1+\delta)}. \quad (1)$$

After comparing their Hölder-regular estimates with the directional Haar estimate, they state the following problem on p. 4 of the arXiv PDF:

1. In (1.9) the interpolation exponents for n depend just on the order α of the Hölder estimates, and not on the value of p .
2. By contrast in (1.9), the $P^{(\varepsilon)}$ estimates show a critical transition at $p = 2$. The exponents in (1.9) are best possible, as shown in [14] Section 8. Hence (1.9) does not arise as the limit of $\alpha \rightarrow 0$ from the estimates (1.8).
3. As Hölder estimates are not available for the Haar system we exploit in [14] that the discontinuities of Haar functions are concentrated at an $(n - 1)$ dimensional set, and that

$$\int_{\mathbb{R}^n} |h_Q^{(\varepsilon)}(x - s) - h_Q^{(\varepsilon)}(x)|^p dx \leq C|s| \cdot |Q|^{\frac{n-1}{n}}. \quad (1.10)$$

4. It remains an open problem to prove interpolatory estimates for directional projections $W^{(\varepsilon)}$ when the underlying wavelets satisfy decay estimates *only*. The particular interest in this question comes from theorems of G. Gripenberg [10] and P. Wojtaszczyk [23] who proved that wavelets with decay (1.2a) form an unconditional basis in L^p , $(1 < p < \infty)$ – *without* using assumptions on smoothness.

Outlook. In [19, 14], proving the weak semi-continuity of separately convex functionals –as

Figure 1: Full-width source crop containing the complete decay-only problem.

In transcription: “It remains an open problem to prove interpolatory estimates for directional projections $W^{(\varepsilon)}$ when the underlying wavelets satisfy decay estimates only.” We show that decay alone is insufficient. The obstruction is already present in $L^2(\mathbb{R}^2)$.

2 Definitions and target class of estimates

Let R_1 be the first Riesz transform on $L^2(\mathbb{R}^2)$, with Fourier multiplier

$$-i \frac{\xi_1}{(\xi_1^2 + \xi_2^2)^{1/2}}.$$

An interpolatory estimate of the source's type has the property that its right-hand side is $o(\|u\|_2)$ whenever $\|R_1 u\|_2 = o(\|u\|_2)$. To cover both the power and logarithmic estimates, we consider the general form

$$\|W^{(1,0)}u\|_2 \leq C\|u\|_2 \Psi\left(\frac{\|R_1 u\|_2}{\|u\|_2}\right), \quad u \neq 0, \quad (2)$$

where $\Psi : [0, 1] \rightarrow [0, \infty)$ satisfies $\Psi(t) \rightarrow 0$ as $t \downarrow 0$. The source's power controls use $\Psi(t) = t^\beta$ with $\beta > 0$, while its Lipschitz endpoint has, up to a constant, $\Psi(t) = t(1 + \log(1/t))$.

3 Proof intuition

Once every direction-specific condition is removed, decay sees where a wavelet is located but not which coordinate its orientation label describes. We exploit this literally: swap the labels of the horizontal and vertical tensor Haar wavelets. This preserves the complete decay-only hypothesis, but the projection now called “(1,0)-directional” actually selects wavelets which cancel in the second coordinate.

A long horizontal strip is a finite sum of those selected wavelets, so the relabeled projection fixes it. As the strip becomes longer, its first-coordinate Fourier frequencies concentrate near zero, whereas its second-coordinate Haar oscillation stays at unit scale. The first Riesz multiplier therefore tends to zero on its Fourier mass. This contradicts every estimate whose control vanishes with the first Riesz norm.

4 Counterexample

For a dyadic interval I , let

$$h_I^0 = \mathbf{1}_I, \quad h_I^1 = \mathbf{1}_{I_L} - \mathbf{1}_{I_R},$$

where I_L, I_R are its two dyadic children. If $Q = I_1 \times I_2$ is a dyadic square and $\eta = (\eta_1, \eta_2) \in \mathcal{A} := \{(1, 0), (0, 1), (1, 1)\}$, put

$$h_Q^\eta(x_1, x_2) = h_{I_1}^{\eta_1}(x_1) h_{I_2}^{\eta_2}(x_2).$$

The family $\{h_Q^\eta/|Q|^{1/2}\}$ is the usual tensor Haar orthonormal basis of $L^2(\mathbb{R}^2)$.

Let π be the permutation of $\mathcal{A}$ which swaps $(1, 0)$ and $(0, 1)$ and fixes $(1, 1)$. Define a relabeled wavelet system by

$$\varphi_Q^\varepsilon = h_Q^{\pi(\varepsilon)}. \quad (3)$$

Relabeling preserves orthonormality and completeness. Moreover,

$$|\varphi_Q^\varepsilon(x)| \leq \mathbf{1}_Q(x) \leq \left(1 + \frac{\text{dist}(x, Q)}{s(Q)}\right)^{-2(1+\delta)}$$

for every $\delta > 0$. Thus (1) holds with $C = 1$, while no regularity or direction-specific cancellation is being assumed.

Let $W^{(1,0)}$ be the directional projection associated with the relabeled system. By (3), it is exactly the ordinary Haar projection $P^{(0,1)}$.

Theorem 1 (Decay-only directional interpolation fails). *For the decay-only wavelet system (3), there are nonzero finite Haar sums $u_N \in L^2(\mathbb{R}^2)$ such that*

$$W^{(1,0)}u_N = u_N, \quad \frac{\|R_1u_N\|_2}{\|u_N\|_2} \longrightarrow 0. \quad (4)$$

Consequently, for this system there are no $C < \infty$ and no function Ψ with $\Psi(t) \rightarrow 0$ for which (2) holds for all $u \neq 0$. In particular, all positive-power interpolatory estimates and the logarithmic endpoint estimate of the source fail under decay only.

Proof. Let

$$a = \mathbf{1}_{[0,1)}, \quad b = \mathbf{1}_{[0,1/2)} - \mathbf{1}_{[1/2,1)},$$

and, for a positive integer N , set

$$u_N(x_1, x_2) = \mathbf{1}_{[0,N)}(x_1)b(x_2) = a(x_1/N)b(x_2). \quad (5)$$

For $0 \leq k < N$, the square $Q_k = [k, k+1) \times [0, 1)$ is dyadic and

$$h_{Q_k}^{(0,1)}(x_1, x_2) = \mathbf{1}_{[k,k+1)}(x_1)b(x_2).$$

Therefore

$$u_N = \sum_{k=0}^{N-1} h_{Q_k}^{(0,1)} = \sum_{k=0}^{N-1} \varphi_{Q_k}^{(1,0)},$$

so $W^{(1,0)}u_N = u_N$ exactly.

It remains to estimate the first Riesz transform. With a unitary Fourier transform, the dilation in (5) gives

$$\widehat{u_N}(\xi_1, \xi_2) = N\widehat{a}(N\xi_1)\widehat{b}(\xi_2), \quad \|u_N\|_2^2 = N\|a\|_2^2\|b\|_2^2 = N.$$

Plancherel and the change of variable $\zeta = N\xi_1$ now yield

$$\frac{\|R_1u_N\|_2^2}{\|u_N\|_2^2} = \int_{\mathbb{R}^2} \frac{(\zeta/N)^2}{(\zeta/N)^2 + \xi_2^2} |\widehat{a}(\zeta)|^2 |\widehat{b}(\xi_2)|^2 d\zeta d\xi_2. \quad (6)$$

For almost every (ζ, ξ_2) the multiplier in (6) tends to zero. It is bounded by one, while $|\widehat{a}(\zeta)|^2 |\widehat{b}(\xi_2)|^2$ is integrable. Dominated convergence proves (4).

If (2) held, applying it to u_N and using $W^{(1,0)}u_N = u_N$ would give

$$1 \leq C\Psi\left(\frac{\|R_1u_N\|_2}{\|u_N\|_2}\right).$$

The right side tends to zero, a contradiction. □

5 Interpretation and limitation

The counterexample isolates the structural issue: localization and decay do not make the label $\varepsilon_i = 1$ signify cancellation in coordinate i . Accordingly, decay alone cannot order the projection by R_i . Some label-compatible directional hypothesis is necessary.

The source's full admissibility definition also includes a sectional oscillation estimate (1.2c) in every coordinate with $\varepsilon_i = 1$. The relabeled Haar system fails that condition in the nominal first coordinate. Hence, if the intended problem was to retain (1.2c) and discard only Holder regularity, that narrower problem is not settled here. The theorem is a full negative answer to the words “decay estimates only” and a sharp boundary obstruction to any stronger intended formulation.

6 Verification and novelty boundary

Every constructed function is a finite Haar sum, the projection identity is exact, and the only limiting step is the displayed Plancherel calculation with an integrable dominating function. The failure at $p = 2$ is enough to rule out a decay-only theorem asserted for the full range $1 < p < \infty$.

Cheap run indexes and two bounded literature-search rounds were completed on 22 July 2026 using the source id, exact open-problem sentence, title, authors, and close arXiv variants. Only the source and its published version were located; no later answer or matching counterexample was found. Novelty confidence is moderate, not exhaustive. Human review should prioritize the literal meaning of “decay estimates only” and the direction-label issue.

References

- [1] P. F. X. Muller and S. Muller, *Interpolatory estimates, Riesz transforms and wavelet projections*, Rev. Mat. Iberoam. **32** (2016), no. 4, 1137–1162; arXiv:1409.2143.